



**S.B. JAIN INSTITUTE OF TECHNOLOGY
MANAGEMENT & RESEARCH, NAGPUR**

Post Lab

Aim: To implement the **Round Robin CPU Scheduling Algorithm** and calculate waiting time and turnaround time of processes.

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Aim: To implement the **Round Robin CPU Scheduling Algorithm** and calculate waiting time and turnaround time of processes.

Objectives:

- To understand the concept of time-sharing in CPU scheduling
- To implement Round Robin algorithm using C programming
- To analyze performance using waiting time and turnaround time Requirements

Hardware Requirements:

- Computer system with minimum 4GB RAM
- Processor (Intel/AMD)
- Keyboard and Monitor

Software Requirements:

- Operating System (Windows/Linux)
- C Compiler (GCC / Turbo C)
- Code Editor (VS Code / Notepad++)

Theory:

Round Robin is a **preemptive scheduling algorithm** used in operating systems for **time-sharing environments**. Each process is assigned a fixed time slice known as **time quantum**.

- Processes are executed in a cyclic order.
- If a process is not completed within the given time quantum, it is moved to the end of the queue.
- This ensures fairness and avoids starvation.

Key Terms:

- **Burst Time (BT):** Time required by process for execution
- **Waiting Time (WT):** Time spent in ready queue
- **Turnaround Time (TAT):** Total time from arrival to completion

Code:

```
#include <stdio.h>

int main() {
    int n, tq, i, time = 0, remain;
    int bt[10], rt[10];
    float wt = 0, tat = 0;

    printf("Enter number of processes: ");
    scanf("%d", &n);

    remain = n;

    for(i = 0; i < n; i++) {
        printf("Enter burst time for process %d: ", i+1);
        scanf("%d", &bt[i]);
        rt[i] = bt[i];
    }

    printf("Enter time quantum: ");
    scanf("%d", &tq);

    i = 0;
    while(remain != 0) {
        if(rt[i] <= tq && rt[i] > 0) {
            time += rt[i];
            rt[i] = 0;
            remain--;
            wt += time - bt[i];
            tat += time;
        }
        else if(rt[i] > 0) {
            rt[i] -= tq;
            time += tq;
        }
        }

        if(i == n-1)
            i = 0;
        else
            i++;
    }
}
```

```
printf("\nAverage Waiting Time = %.2f", wt/n);  
printf("\nAverage Turnaround Time = %.2f\n", tat/n);  
  
return 0;  
}
```

Conclusion:

The Round Robin scheduling algorithm was successfully implemented. It provides fair CPU allocation among processes and is suitable for time-sharing systems, though performance depends on the chosen time quantum.

❖ Discussion Questions:

1. What is the significance of the pwd command in a Linux environment?
2. Explain the function of the cp command and its common options.
3. How does chmod 700 affect file permissions, and what does each digit represent?
4. Describe the difference between head and tail commands in Linux.
5. What is the purpose of the grep command, and how is it used with regular expressions?

❖ References:

<https://ubuntu.com/tutorials/command-line-for-beginners#l-overview>
<https://www.geeksforgeeks.org/25-basic-ubuntu-commands/>

Date: ____ / ____ /2026

Signature

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